Janak Kabra

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Education

Vellore Institute of Technology — Vellore, Tamil Nadu
Bachelor of Technology in Computer Science and Engineering — *Aug. 2024 – May 2028*

Experience

Self-Employed — Remote
Web Developer (Freelance) — *Jun. 2025 – Present*

- Led end-to-end SDLC for client applications using React, Vite, and Node.js to construct high-performance, mobile-first web platforms.
- Architected modular, reusable UI components using Tailwind CSS and Framer Motion, reducing codebase redundancy by 40% and boosting delivery speed of features.
- Implemented frontend performance enhancements including asset lazy loading and code-splitting, achieving 95+ Google Lighthouse scores and reducing average load times by 600ms.

Mythus AI Studios — Remote
AI Research Intern — *Oct. 2025 – Dec. 2025*

- Conducted exploratory data analysis (EDA) and data cleansing on heterogeneous unstructured datasets to validate core classification models prior to production deployment.
- Optimized calculation speed of feature selection scripts, reducing pipeline runtimes by 22.4% using vectorized NumPy operations and multiprocessing libraries in Python.
- Translated academic machine learning papers on computer vision and neural networks into functional Python blueprints and documentation, establishing a central knowledge base.

Infosys Springboard — Remote
AI Domain Intern — *Feb. 2026 – Jun. 2026*

- Engineered and benchmarked multivariate regression and LSTM forecasting models for "AgriYield Predictor" to predict crop yield variances with a 14.2% reduction in forecast variance.

Projects

Volatility Surface Engine | GitHub | Live Demo | *Python, Numba, FastAPI* — May 2026

- Built an options pricing, calibration, and risk engine from first principles, integrating a stable Heston stochastic volatility Fourier solver and arbitrage-free SVI implied volatility surface model.
- Resolved numerical instability ("Little Heston Trap") in the characteristic function at long maturities ($T \geq 10$) using branch-cut-free representations of the complex logarithm.
- Calibrated the Heston model parameters by minimizing Black-Scholes Vega-weighted price errors as a first-order implied volatility proxy, running desk-ready calibration in under 100ms.

Deterministic Algo Debugger | GitHub | Live Demo | *React, Zustand, Canvas* — Apr. 2026

- Developed a performance visualizer tracking algorithmic complexity metrics (swap counts, array comparisons) for Quick, Heap, and Merge sort algorithms.
- Engineered a zero-allocation step recorder that runs algorithms in a lightweight virtual container, generating a queue of immutable snapshots for dynamic Big-O analysis.
- Designed a Dual Algorithm Race mode to visualize concurrent thread execution, evaluating execution speeds under identical inputs.

NSE Stock Screener | GitHub | Live Demo | *FastAPI, PostgreSQL, React* — Feb. 2026

- Built a high-performance fundamental stock screener for NIFTY 500 equities, delivering complex multi-factor query results in sub-500ms.
- Developed a peer-relative sector ranking engine, calculating financial percentile statistics (P/E, ROE, Debt/Equity) against industry peers in real-time.
- Designed database indexing strategies and optimized SQL queries to run chained filter combinations in a single database roundtrip, minimizing latency.

Technical Skills

Languages: Python, C++, TypeScript, JavaScript (ES6+), SQL, C, Go, Java, HTML, C#
Frameworks & Libraries: React, Next.js, Node.js, FastAPI, PyTorch, TensorFlow, OpenCV, Zustand, Tailwind CSS, Framer Motion, GSAP, NumPy, Pandas, SciPy, Numba JIT, Plotly
Datastores & Developer Tools: PostgreSQL, MySQL, Git, Docker, Linux, GitHub Actions, Vercel, Postman, Excel, Power BI, Alteryx.